

Sp26 - CHEMICAL ENGINEERING MATERIALS

The University of Texas at Austin — Department of Chemical Engineering

ChE 350: Chemical Engineering Materials

UNIQUE #: 16510

CLASS TIMES: 10-11 a.m. Monday-Wednesday-Friday in CPE 2.216

RECITATION TIMES: 8:30 a.m. – 9:30 a.m. on Tuesday in CPE 2.216

Instructor: Professor Jim Chelikowsky. Office: POB 4.324. Office hours: Monday 1-2 p.m. (or by appointment). Email: jrc@utexas.edu. Phone: 512-323-9083

Teaching Assistant: Seungjun Lee. Office hours: Friday 3-5 p.m. Location: NHB 5.320F. Email: dltmdwns1997@utexas.edu

Course Description

Our course is an introduction to the fundamentals and applications of materials science and engineering. Materials science is about understanding and predicting how materials respond to an outside influence such as temperature, mechanical stress, chemical reagents, electrical and magnetic fields, radiation... and using that understanding to engineer better materials for a variety of applications. We will target the following goals:

- To teach elementary relationships between structure and properties of materials.
- To introduce various classes of materials, including metals, ceramics, polymers, and semiconductors.
- Define and introduce characteristics of materials such as mechanical and electronic properties, including magnetic and optical properties.
- Role of thermodynamics (through phase diagrams) and diffusion play in determining properties.

Prerequisites: Chemical Engineering major or consent of instructor.

Course materials

We will not have a required textbook. I will distribute lecture notes on Canvas and we will follow (more or less) the textbook: Materials Science and Engineering: An Introduction 10th Edition by [William D. Callister Jr.](#) and [David G. Rethwisch](#), but you need not buy a copy. I will post materials on Canvas needed.

Course Outline

All instructions, assignments, readings, and essential information will be on the Canvas website. Lecture notes will be posted on Canvas. *Please be sure to check Canvas regularly.*

Course Requirements and Grading

Coursework: This course will consist of keeping up with reading, homework, three exams, and one final exam.

Course devices:

Please be sure to have a device with access to your Canvas student account when

participating in lectures. We will use Canvas for various polls. It is recommended to have a calculator that is capable of basic arithmetic functions, exponents, logarithms, parentheses.

Course mode:

Face-to-face unless otherwise specified. Lectures will be recorded and made

available on Canvas as announced.

GRADES

Coursework Breakdown:

- Homework (15 %): However, you will be allowed to drop two assignments. Solutions will need to be scanned and uploaded to Canvas.
- Participation (10 %): Response to polls and other activities in lecture or recitation. To receive full credit you need to participate in at least half the polls.
- Mid-term exams in class total 45 %, 15% each; Final Exam 30%.

Grade breakdown: This is *approximate*, but we will try to use this scale. > 88% (A), 88-87% (A-), 82-87% (B+), 75-82% (B), 71-75% (B-), 68-71% (C+), 58-68% (C), 55-58% (C-), 50-55% (D), < 50% (F)

COURSE TIPS FOR SUCCESS

Regular Participation: Lecture, office hours, and recitation are all opportunities for you to interact with the course material and with the instructor and TA. We are here to help you learn and succeed. Please ask questions and ask frequently! Communicate any anticipated and reasonable absences. The instructor and/or TA can work with you to make sure you keep up with the course.

Keep up with the material: These include but are not limited to attending lectures regularly, reading the assigned material, doing the homework assignments and checking Canvas regularly.

Enjoy the learning process: Materials science is a rapidly expanding and exciting field. Most everything you use in your daily lives is made up of “materials.” Few technological problems cannot be solved using a better material!

LECTURE TOPICS

Class Topic	Reading (Callister)
Introduction to Materials	Ch 1
Atomic Structure and Interatomic Interactions	Ch 2
Crystallographic systems, directions, planes	Ch 3
XRD and Diffraction	Ch 3
Crystallographic defects	Ch 4
Midterm Exam on Wednesday, February 11	
Diffusion in solids	Ch 5
Mechanical Properties of metals and ceramics	Ch 6
Materials failure- fracture	Ch 7 & 8
Phase diagrams and phase transformations	Ch 9
Midterm Exam on Wednesday, March 11	
Ceramics	Ch 12
Polymers: structure, properties, and synthesis	Ch 14
Electronic properties	Ch 18
Semiconductors	Ch 18
Midterm Exam on Wednesday, April 15	
Magnetic Properties and Materials	Ch 20
Optical properties of materials	Ch 21

Final Exam TBA	
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OFFICE HOURS

During office hours you are welcome to discuss any topics on materials whether or not currently being discussed in class. Please come by! Office hours may be in-person or remote.

ACADEMIC INTEGRITY EXPECTATIONS

You are expected to uphold all standards of academic integrity described in the University Code of Conduct and the Student Honor Code. Students who violate University rules on academic dishonesty are subject to disciplinary penalties, including the possibility of failure in the course and/or dismissal from the University. Since such dishonesty harms the individual, all students, and the integrity of the University, policies on academic dishonesty will be strictly enforced.

For further information, please visit the Student Conduct and Academic Integrity website at: <http://deanofstudents.utexas.edu/conduct>.

SHARING OF COURSE MATERIALS IS PROHIBITED

No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class without explicit, written permission of the instructor. Unauthorized sharing of materials promotes cheating. The University is well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure of the course.

GETTING HELP WITH TECHNOLOGY

Students needing help with technology in this course should contact the ITS Service Desk.

University Resources for Students

SERVICES FOR STUDENTS WITH DISABILITIES (SSD)

I am committed to creating an accessible and inclusive learning environment in this class. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think

you may have a disability, and need accommodations please contact Services for Students with Disabilities (SSD). Please refer to SSD's website for contact and more information: <http://diversity.utexas.edu/disability/>. If you are already registered with SSD, please discuss with and/or deliver your Accommodation Letter to Prof. Chelikowsky at the start of the semester your accommodations and needs in this course.

COUNSELING AND MENTAL HEALTH CENTER (CMHC)

If you or anyone you know is experiencing symptoms of stress, anxiety, depression, academic concerns, loneliness, difficulty sleeping, or any other concern impacting your well-being – you are strongly encouraged to connect with CMHC. The Counseling and Mental Health Center provides a wide variety of mental health services to all UT students including crisis services, counseling services with immediate support and well-being resources. Additionally, CARE Counselors are located within the academic schools and colleges. These counselors get to know the concerns that are unique to their college's students. For more information on CMHC, visit <https://cmhc.utexas.edu> or call 512-471-3515.

Important Safety Information

If you have concerns about the safety or behavior of fellow students, TAs or professors, contact BCCAL (the Behavior Concerns and COVID-19 Advice Line) at <https://safety.utexas.edu/behavior-concerns-advice-line> or by calling 512-232-5050. Confidentiality will be maintained as much as possible, however the university may be required to release some information to appropriate parties.

CLASSROOM SAFETY AND COVID-19

To help preserve our in-person learning environment, please consider the following:

- Adhere to university mask guidance. Masks are strongly recommended inside university buildings for vaccinated and unvaccinated individuals.
- Vaccinations are widely available, free and not billed to health. The vaccine will help protect against the transmission of the virus to others and reduce serious symptoms in those who are vaccinated.
- Proactive Community Testing remains an important part of the university's efforts to protect our community. Tests are fast and free.
- We encourage the use of the Protect Texas App each day prior to coming to
- If you develop COVID-19 symptoms or feel sick, stay home and contact the University Health Services' Nurse Advice Line at 512-475-6877. If you need to be absent from class, contact Student Emergency Services and they will notify your professors. In addition, to help understand what to do if you have been had close contact with someone who tested positive for COVID-19, see University Health Services link.
- BCCAL remains available as the primary tool to address questions or concerns from the university community about COVID-19.
- Students who test positive should contact BCCAL or self-report (if tested off campus) to University Health Services. Visit Protect Texas Together for more

TITLE IX DISCLOSURE

Beginning January 1, 2020, Texas Senate Bill 212 requires all employees of Texas universities, including faculty, to report any information to the Title IX Office regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu.

For more information about reporting options and resources, visit <http://www.titleix.utexas.edu/> contact the Title IX Office via email at titleix@austin.utexas.edu, or call 512-471-0419.

Although graduate teaching and research assistants are not subject to Texas Senate Bill 212, they are still mandatory reporters under Federal Title IX laws and are required to report a wide range of behaviors we refer to as sexual misconduct, including the types of sexual misconduct covered under Texas Senate Bill 212. The Title IX office has developed supportive ways to respond to a survivor and compiled campus resources to support survivors.

Faculty members are considered “mandatory reporters,” which means that they are required to report violations of Title IX to the Title IX Coordinator. **As such, they must report any Title IX-related incidents** that are disclosed in writing, discussion, or one-on-one. Before talking with me or with any faculty or staff member about a Title IX-related incident, be sure to ask whether they are mandatory reporters. If you want to speak with someone for support or remedies *without making an official report* to the university, email advocate@austin.utexas.edu For more information about reporting options and resources, visit the Title IX Office or email titleix@austin.utexas.edu.